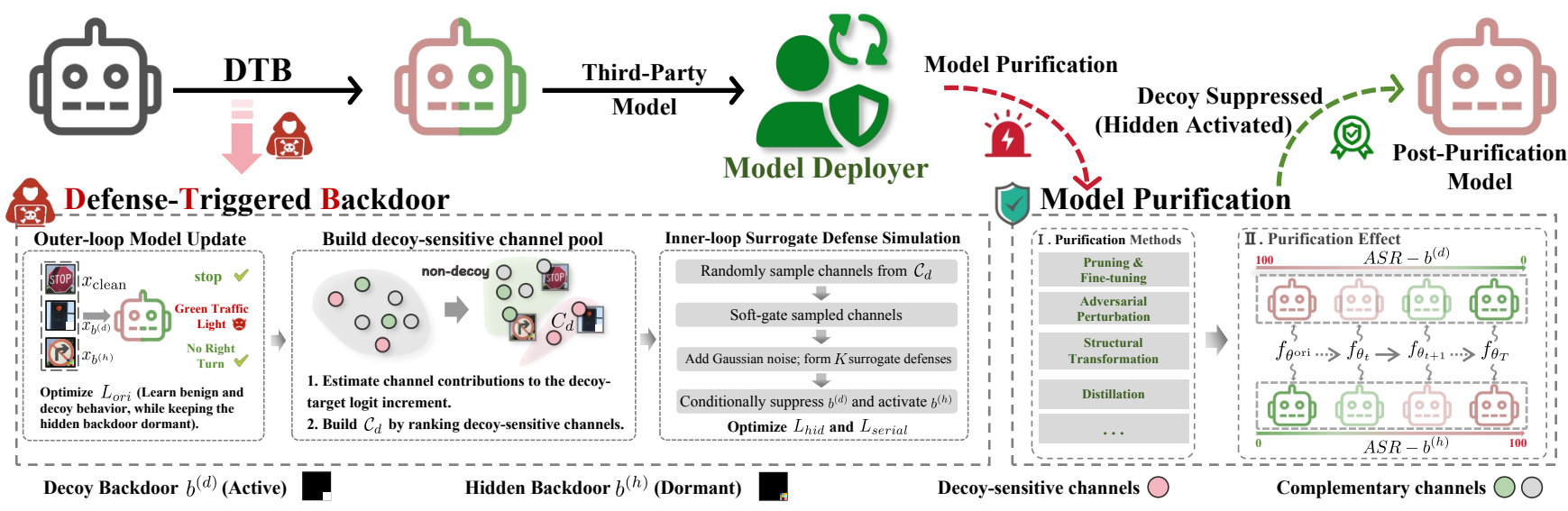

DTB
Third-Party Model
Model Deployer
Model Purification
Decoy Suppressed (Hidden Activated)
Post-Purification Model
Defense-Triggered Backdoor
Outer-loop Model Update
stop
Green Traffic Light
No Right Turn
Optimize L_{ori} (Learn benign and decoy behavior, while keeping the hidden backdoor dormant).
Build decoy-sensitive channel pool
non-decoy
1. Estimate channel contributions to the decoy-target logit increment.
2. Build $\mathcal{C}_d$ by ranking decoy-sensitive channels.
Inner-loop Surrogate Defense Simulation
Randomly sample channels from $\mathcal{C}_d$
Soft-gate sampled channels
Add Gaussian noise; form K surrogate defenses
Conditionally suppress $b^{(d)}$ and activate $b^{(h)}$
Optimize L_{hid} and L_{serial}
Model Purification
I. Purification Methods
Pruning & Fine-tuning
Adversarial Perturbation
Structural Transformation
Distillation
...
II. Purification Effect
$ASR - b^{(d)}$
$ASR - b^{(h)}$
Decoy Backdoor $b^{(d)}$ (Active)
Hidden Backdoor $b^{(h)}$ (Dormant)
Decoy-sensitive channels
Complementary channels